

A Counterexample to Remark 3.3 in arXiv:1612.03808

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Status

This packet gives a likely valid counterexample to the question in Remark 3.3 of Prochazka and Rueda Zoca, *A characterisation of octahedrality in Lipschitz-free spaces*, arXiv:1612.03808. The source paper asks whether assertion (3) in that remark implies assertion (1). The answer is negative: for $r = 2/5$, the metric space already appearing in Example 3.6 satisfies assertion (3), in the only meaningful range $0 < \varepsilon < 1 - r$, but fails assertion (1).

Source Passage

Remark 3.3 states a chain of implications between four assertions, indexed by $0 \leq r < 1$, and says that it is unknown whether assertion (3) implies assertion (1).

which finishes the proof. ■

Remark 3.3. Let M be a metric space and $0 \leq r < 1$. Note that, adapting the proof of Theorem 3.1, it can be proved that each of the following assertion implies the next one:

- (1) For every $\mu_1, \dots, \mu_n \in S_{\mathcal{F}(M)}$ and every $\varepsilon > 0$ there exists $u \neq v \in M$ such that

$$\left\| \mu_i + \frac{\delta_u - \delta_v}{d(u, v)} \right\| \geq 2 - r - \varepsilon$$

holds for every $i \in \{1, \dots, n\}$.

- (2) For each finite subset $N \subseteq M$ and $\varepsilon > 0$, there exist $u, v \in M, u \neq v$, such that

$$(1 - r - \varepsilon)(d(x, y) + d(u, v)) \leq d(x, u) + d(y, v)$$

holds for all $x, y \in N$.

- (3) For each $\varepsilon > 0$ and each finite subset $N \subset M$ there are points $u, v \in M, u \neq v$, such that every 1-Lipschitz function $f : N \rightarrow \mathbb{R}$ admits an extension $\tilde{f} : M \rightarrow \mathbb{R}$ which is $\frac{1}{1-r-\varepsilon}$ -Lipschitz and satisfies $\tilde{f}(u) - \tilde{f}(v) \geq d(u, v)$.

- (4) For every $\mu_1, \dots, \mu_n \in S_{\mathcal{F}(M)}$ and every $\varepsilon > 0$ there exists $u \neq v \in M$ such that

$$\left\| \mu_i + \frac{\delta_u - \delta_v}{d(u, v)} \right\| \geq 2 - 2r - \varepsilon$$

holds for every $i \in \{1, \dots, n\}$.

We do not know whether (3) actually implies (1). Moreover, notice that Theorem 3.1 is the particular case of the above implications whenever $r = 0$. Finally, notice that assertion (1) is equivalent to the fact that the *Whitley's thickness index of $\mathcal{F}(M)$* is greater than or equal to $2 - r$ (we refer to [5] and references therein for formal definitions and background on such index).

It is time to give some examples of metric spaces enjoying the LTP. Notice that Proposition 4.7 provides a sweeping generalization of the point (3) below but we postpone it till Section 4 as its proof requires some preparatory work.

For reference, the two assertions relevant here are:

- (1) For every $\mu_1, \dots, \mu_n \in S_{\mathcal{F}(M)}$ and every $\varepsilon > 0$, there is $u \neq v \in M$ such that

$$\left\| \mu_i + \frac{\delta_u - \delta_v}{d(u, v)} \right\| \geq 2 - r - \varepsilon$$

for all i .

- (3) For each $\varepsilon > 0$ and each finite $N \subset M$, there are $u, v \in M, u \neq v$, such that every 1-Lipschitz function $f : N \rightarrow \mathbb{R}$ admits an extension $\tilde{f} : M \rightarrow \mathbb{R}$ which is $(1 - r - \varepsilon)^{-1}$ -Lipschitz and satisfies

$$\tilde{f}(u) - \tilde{f}(v) \geq d(u, v).$$

The denominator in (3) forces the usual small- ε reading $0 < \varepsilon < 1 - r$. For larger ε the displayed Lipschitz constant is not positive, so the original statement is not a meaningful metric extension assertion.

Proof Intuition

The relaxed extension assertion only needs two new points u, v on which one can force a unit drop while remaining within a fairly large Lipschitz constant. In the source paper's Example 3.6 metric, there are infinitely many points x_n all at distance one from every old point except themselves, so after a finite set N is chosen we can pick two fresh points x_i, x_j . Because the whole metric has diameter two, every 1-Lipschitz function on N oscillates by at most two. For $r = 2/5$, the permitted constant is larger than $3/2$, exactly enough room to assign values a and $a - 1$ to the two fresh points and then extend.

The free-space thickness assertion is stricter. Testing only the two opposite molecules $\mu = (\delta_0 - \delta_z)/2$ and $-\mu$ already blocks it: every candidate molecule points too close to at least one of these two directions. The Arens-Eells decompositions below show a uniform upper bound $3/2$, while assertion (1) would demand values arbitrarily close to $2 - r = 8/5$.

The Counterexample

Let

$$M = \{0, z\} \cup \{x_n : n \in \mathbb{N}\}$$

with

$$d(0, z) = 2, \quad d(a, b) = 1$$

for every other pair of distinct points. This is a metric space; it is the same metric used in Example 3.6 of the source paper.

Theorem 1. *For $r = 2/5$, assertion (3) of Remark 3.3 holds for M , but assertion (1) fails for M . Hence assertion (3) does not imply assertion (1).*

Proof. First prove assertion (3). Let $N \subset M$ be finite and let $0 < \varepsilon < 1 - r = 3/5$. Put

$$L = \frac{1}{1 - r - \varepsilon} = \frac{1}{3/5 - \varepsilon}.$$

Then $L > 5/3 > 3/2$. Choose distinct $u = x_i$ and $v = x_j$ outside N . Thus $d(u, v) = 1$ and $d(u, s) = d(v, s) = 1$ for every $s \in N$.

Let $f : N \rightarrow \mathbb{R}$ be 1-Lipschitz. Since $\text{diam } M = 2$, the oscillation $\text{osc}_N f = \max_N f - \min_N f$ is at most 2. We need a real number a such that

$$|a - f(s)| \leq L, \quad |a - 1 - f(s)| \leq L \quad (s \in N).$$

The simultaneous conditions are equivalent to

$$a \in [\max_N f + 1 - L, \min_N f + L].$$

This interval is nonempty because

$$\max_N f + 1 - L \leq \min_N f + L \iff 1 + \text{osc}_N f \leq 2L,$$

and $1 + \text{osc}_N f \leq 3 < 2L$. Choose such an a , and define

$$g|_N = f, \quad g(u) = a, \quad g(v) = a - 1.$$

The preceding inequalities control the pairs joining N to u or v ; on N itself the map is L -Lipschitz because $L > 1$, and on the pair $\{u, v\}$ because $|g(u) - g(v)| = 1 \leq Ld(u, v)$. Hence g is L -Lipschitz on $N \cup \{u, v\}$, and $g(u) - g(v) = 1 = d(u, v)$. The standard formula

$$\tilde{g}(p) = \inf_{q \in N \cup \{u, v\}} (g(q) + Ld(p, q))$$

extends g to an L -Lipschitz function on all of M . Therefore assertion (3) holds.

Now prove that assertion (1) fails. Let

$$\mu = \frac{\delta_0 - \delta_z}{d(0, z)} = \frac{\delta_0 - \delta_z}{2} \in S_{\mathcal{F}(M)}.$$

We will show that for every molecule

$$m_{uv} = \frac{\delta_u - \delta_v}{d(u, v)}$$

one has

$$\min\{\|\mu + m_{uv}\|, \|-\mu + m_{uv}\|\} \leq \frac{3}{2}.$$

This uses only the elementary upper estimate for the Arens-Eells norm: if $\eta = \sum_k a_k(\delta_{p_k} - \delta_{q_k})$ with $a_k \geq 0$, then

$$\|\eta\| \leq \sum_k a_k d(p_k, q_k).$$

If $\{u, v\} = \{0, z\}$, then m_{uv} is μ or $-\mu$, so one of $\mu + m_{uv}$, $-\mu + m_{uv}$ is zero. If exactly one endpoint is one of $\{0, z\}$ and the other is some x_n , one of the two norms is at most 1. For instance,

$$-\mu + m_{0x_n} = \frac{1}{2}(\delta_z - \delta_{x_n}) + \frac{1}{2}(\delta_0 - \delta_{x_n}),$$

and both displayed distances are 1; the other orientations are identical by symmetry.

It remains to consider $u = x_i$, $v = x_j$, $i \neq j$. Then

$$\mu + m_{uv} = \frac{1}{2}(\delta_0 - \delta_v) + \frac{1}{2}(\delta_u - \delta_z) + \frac{1}{2}(\delta_u - \delta_v),$$

and

$$-\mu + m_{uv} = \frac{1}{2}(\delta_z - \delta_v) + \frac{1}{2}(\delta_u - \delta_0) + \frac{1}{2}(\delta_u - \delta_v).$$

All distances appearing in these two decompositions are 1, so both norms are at most $3/2$. Hence the displayed minimum is always at most $3/2$.

For $r = 2/5$, assertion (1) would require, for the pair $\mu, -\mu$, that there exist a molecule m_{uv} with

$$\|\mu + m_{uv}\| \geq 2 - r - \varepsilon, \quad \|-\mu + m_{uv}\| \geq 2 - r - \varepsilon.$$

Taking $\varepsilon = 1/20$, this lower bound is

$$2 - \frac{2}{5} - \frac{1}{20} = \frac{31}{20} > \frac{3}{2},$$

contradicting the uniform estimate above. Thus assertion (1) fails. $\square$

Computational Check

The script `code/verify_norm_bounds.py` solves the finite Arens-Eells transport linear programs for the four representative points $\{0, z, a, b\}$. It confirms

$$\max_{u \neq v} \min \{ \|\mu + m_{uv}\|, \|-\mu + m_{uv}\| \} = 3/2.$$

The proof above does not depend on this computation; it is included as an independent sanity check of the molecule case analysis.

Novelty and Literature Check

The run indexes `registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, and `proof_gaps/index.tsv` were searched for the arXiv id, title phrases, Remark 3.3, thick3, thick1, Whitley’s thickness index, and long trapezoid property phrases. No prior packet for this target was found.

External web searches on 2026-06-20 for exact and near-exact phrases included:

- “We do not know whether” with “thick3” and “thick1”;
- “assertion (3)” and “assertion (1)” with “Lipschitz-free”;
- “Whitley’s thickness index” with “Lipschitz-free” and “long trapezoid”.

These searches returned the source paper or no answer to Remark 3.3. The novelty confidence is therefore good but still subject to human literature review.

References

References

- [1] A. Prochazka and A. Rueda Zoca, *A characterisation of octahedrality in Lipschitz-free spaces*, arXiv:1612.03808, 2016.

Human Review Recommendation

The recommended review focus is the small- ε convention in assertion (3). Since the source’s constant $(1 - r - \varepsilon)^{-1}$ is positive only for $\varepsilon < 1 - r$, the counterexample is complete under the necessary reading of the statement.